COM 3401
DATA COMMUNICATION AND NETWORKING

Lecture 10

FLOW CONTROL

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Topics covered

- Introduction to Flow Control
- Stop-and-Wait Flow Control
- Sliding-Window Flow Control

Reference:

William Stallings, *Data and Computer Communications*, 8th Edition

Andrew S. Tanenbaum, *Computer Networks,* 5th Edition

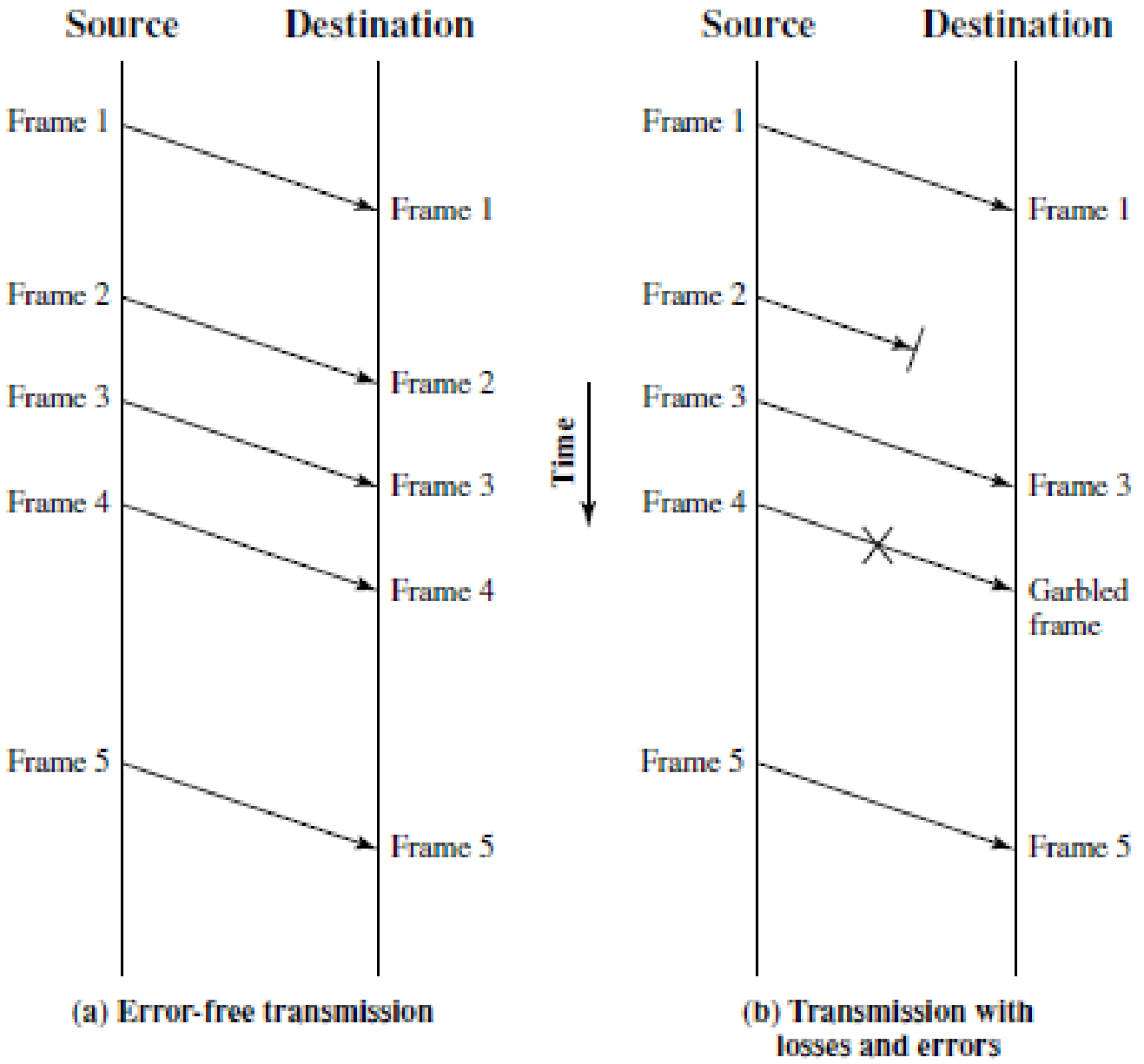
10.1 Introduction to Flow Control



Flow Control

- A mechanism that prevents a transmitting station from sending data faster than the receiving station can absorb it.
- **Main purpose**
  - Protect the receiver from **buffer overflow**
  - Regulate the rate of frame transmission
  - Coordinate the sender and receiver
- **Two techniques**
  1. **Stop-and-Wait Flow Control**
  2. **Sliding-Window Flow Control**

Flow Control



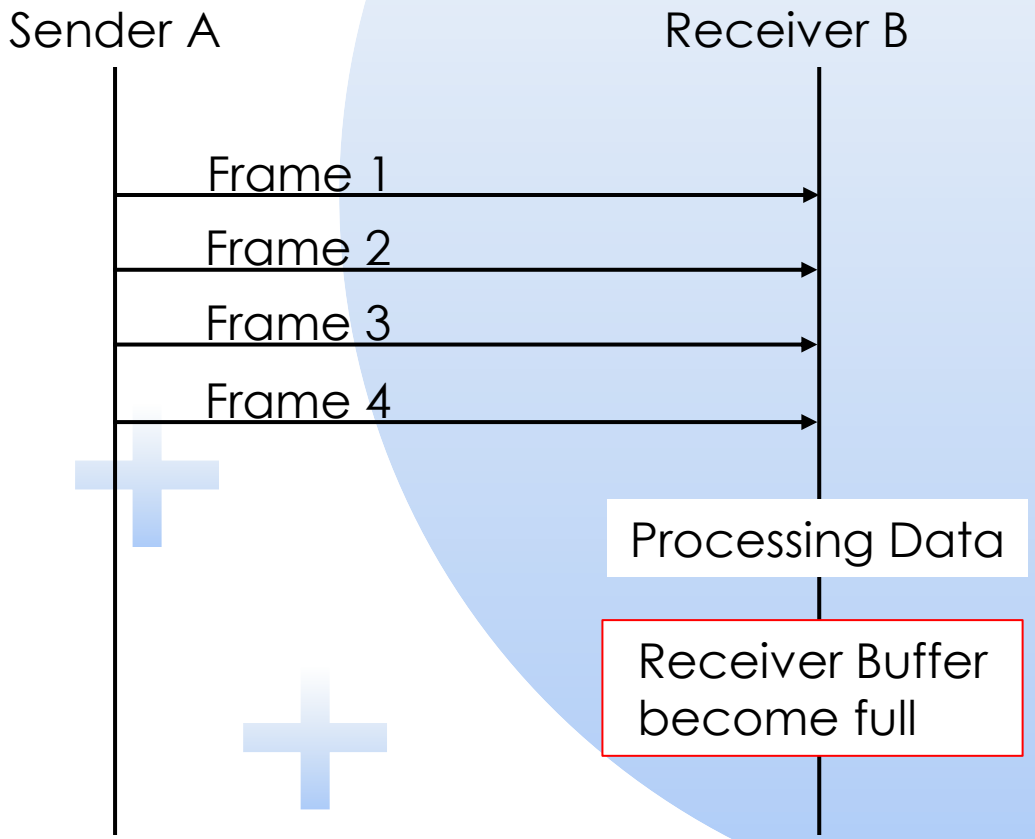
(a) Error-free transmission

(b) Transmission with losses and errors

Why Is Flow Control Needed?

Without flow control

- **Sender rate > Receiver processing rate**
- Results in:
  - Frames accumulate
  - Receiver buffer may overflow
  - Data can be lost



10.2 Stop-and-Wait Flow Control



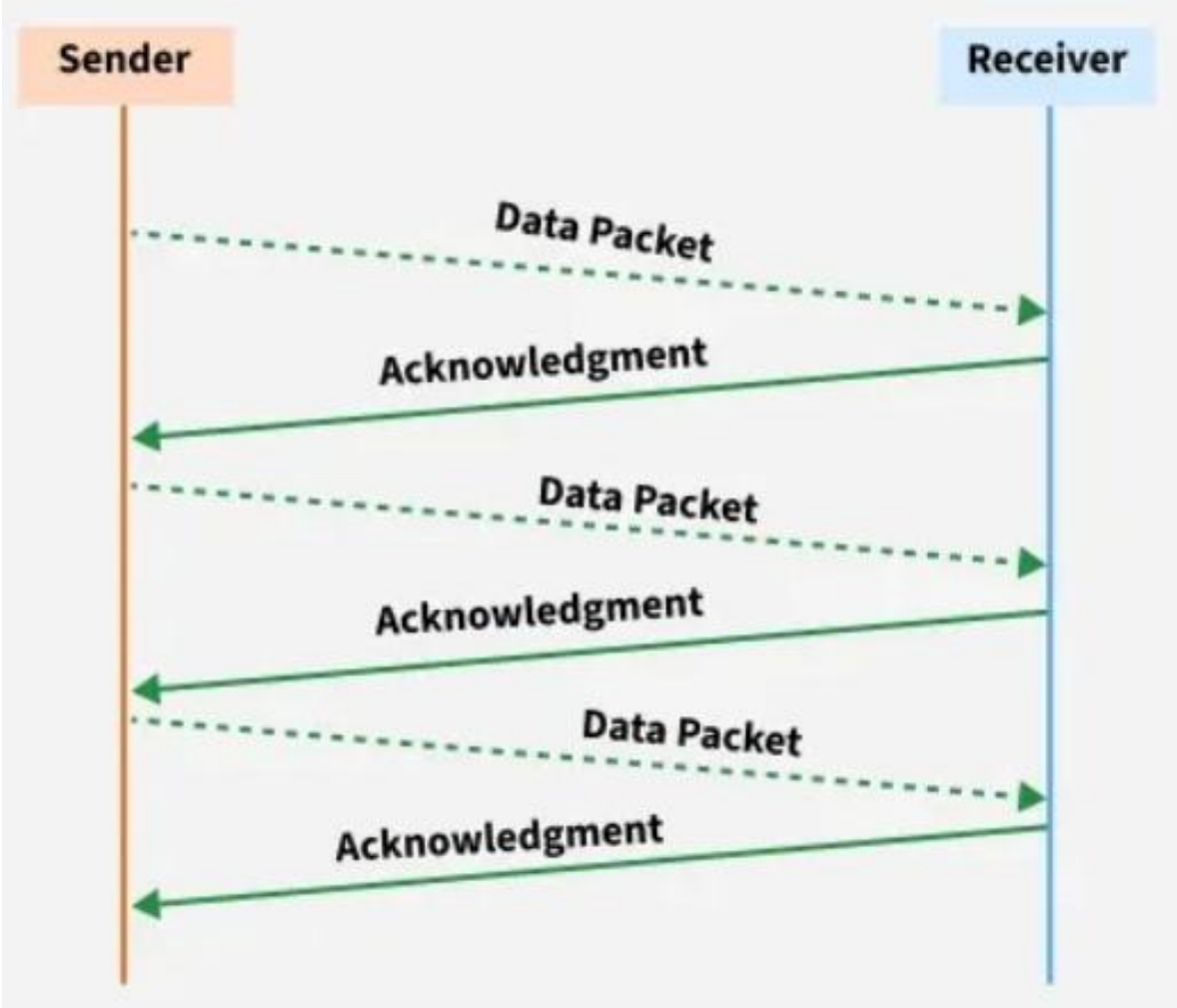
Stop-and-Wait Flow Control

Basic idea:

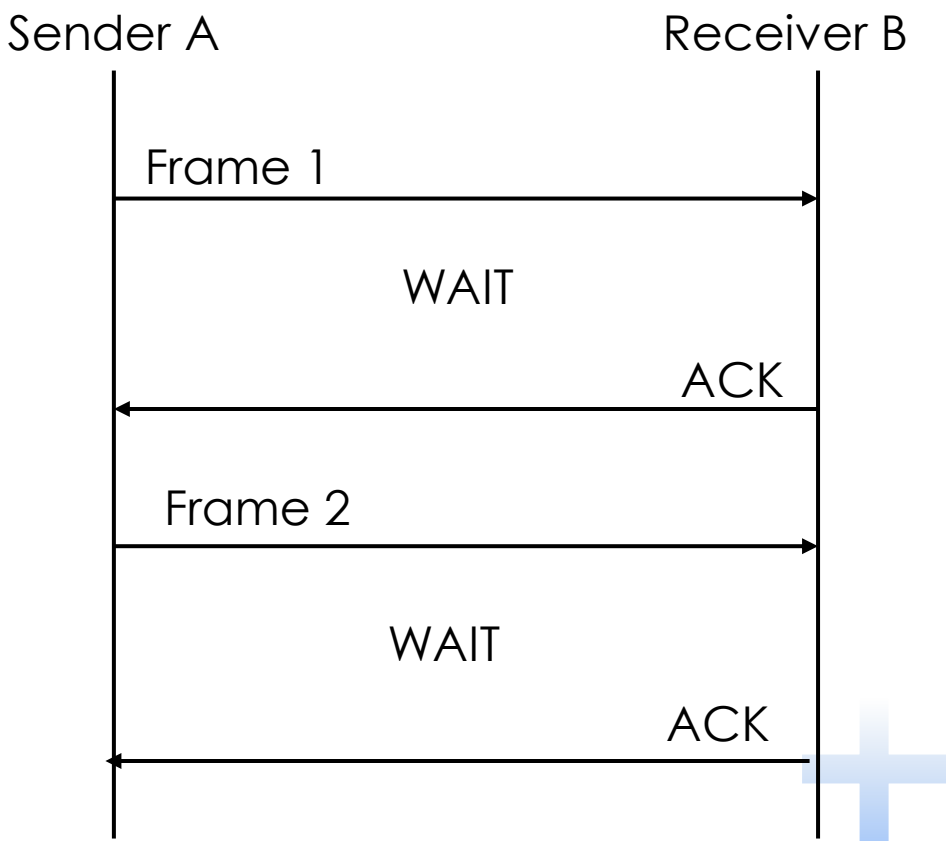
Send **one frame**, then wait before sending the next frame.

Operation

1. Sender transmits one frame.
2. Sender stops transmitting.
3. Receiver receives and processes the frame.
4. Receiver sends an acknowledgment.
5. Sender sends the next frame.



Stop-and-Wait: Sender and Receiver



Key characteristic

- **Only one frame can be outstanding at a time.**

Stop-and-Wait and Link Utilization

- The sender spends time:
  - Transmitting the frame
  - Waiting for the frame to reach the receiver
  - Waiting for the acknowledgment
  - Waiting for the acknowledgment to return
- Therefore:
  - **Much of the link can remain idle.**
- **Main problem**
  - **Long propagation delay → more waiting → lower utilization**

Stop-and-Wait: Efficiency Idea

Normalized propagation delay

$$a = \frac{t_{prop}}{t_{frame}}$$

where:

- t_{prop} =propagation time
- t_{frame} =frame transmission time

Stop-and-wait link utilization:

$$U = \frac{1}{1 + 2a}$$

Therefore

If propagation time becomes large compared with frame transmission time:
a↑ ⇒ U↓

Stop-and-Wait Example: Satellite Link

Consider:

- 1 Mbps satellite link
- 36,000 km geosynchronous satellite altitude
- Frame = **8000 bits**
- Frame transmission time = **8 ms**
- Approximate round-trip waiting produces a total cycle of **488 ms**

Therefore:

- Frame transmission = 8 ms
- Total cycle = 488 ms

Only a small fraction of the cycle is actually used to transmit data.

Key observation:

- **Long propagation delay makes stop-and-wait highly inefficient.**

10.3 Sliding-Window Flow Control



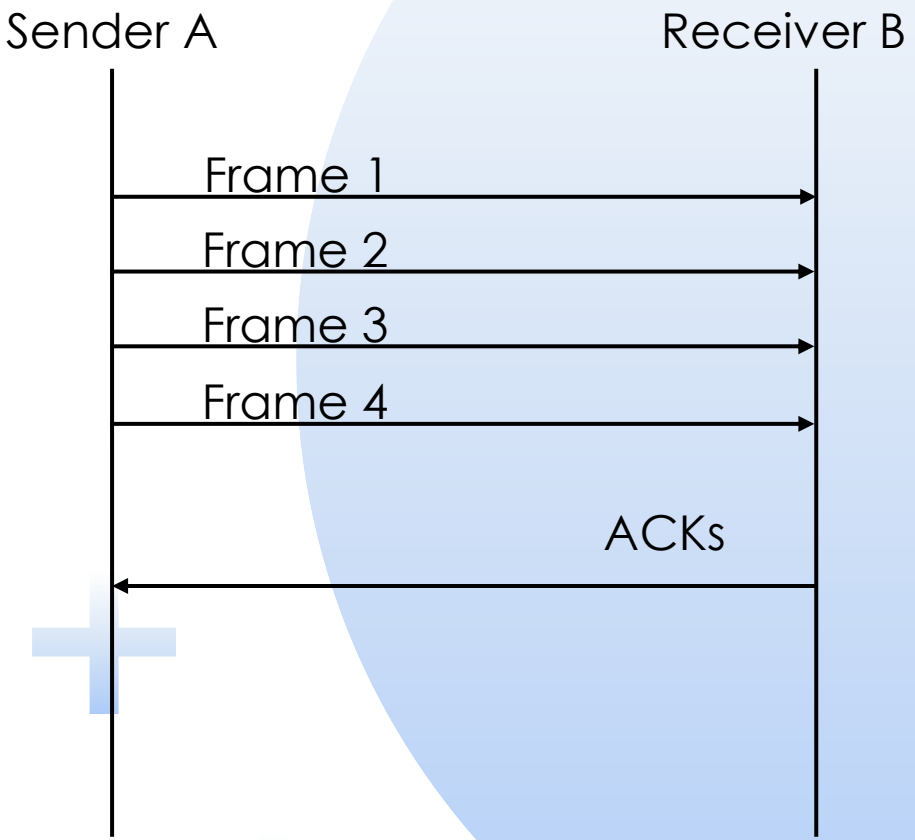
Why Sliding-Window Flow Control?

Problem with Stop-and-Wait

- Only one frame can be in transit.

Solution

- Allow **multiple frames to be in transit simultaneously**.



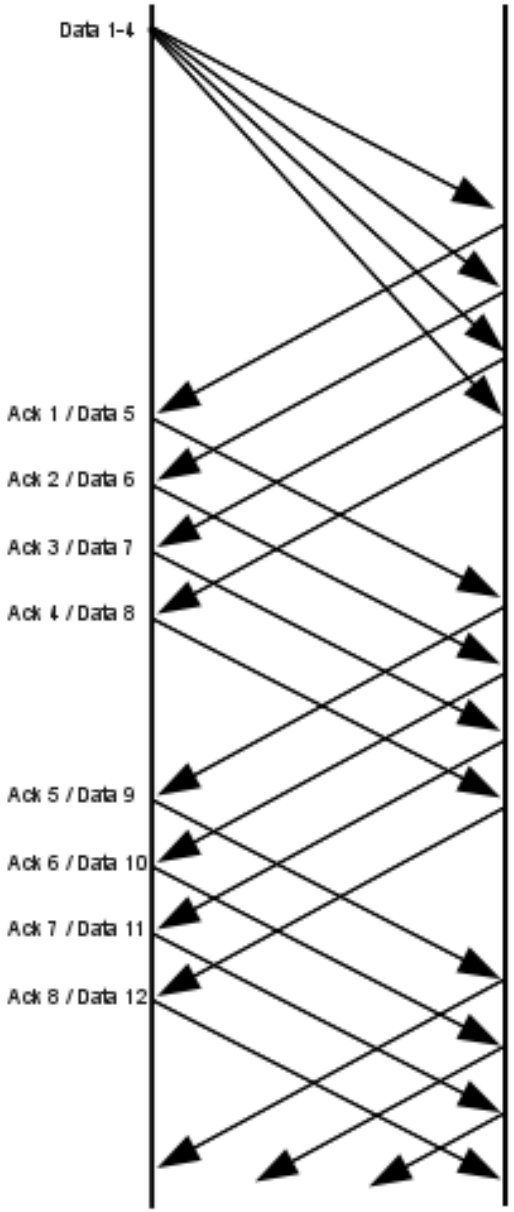
Key idea

- **Keep the link busy while waiting for acknowledgments.**

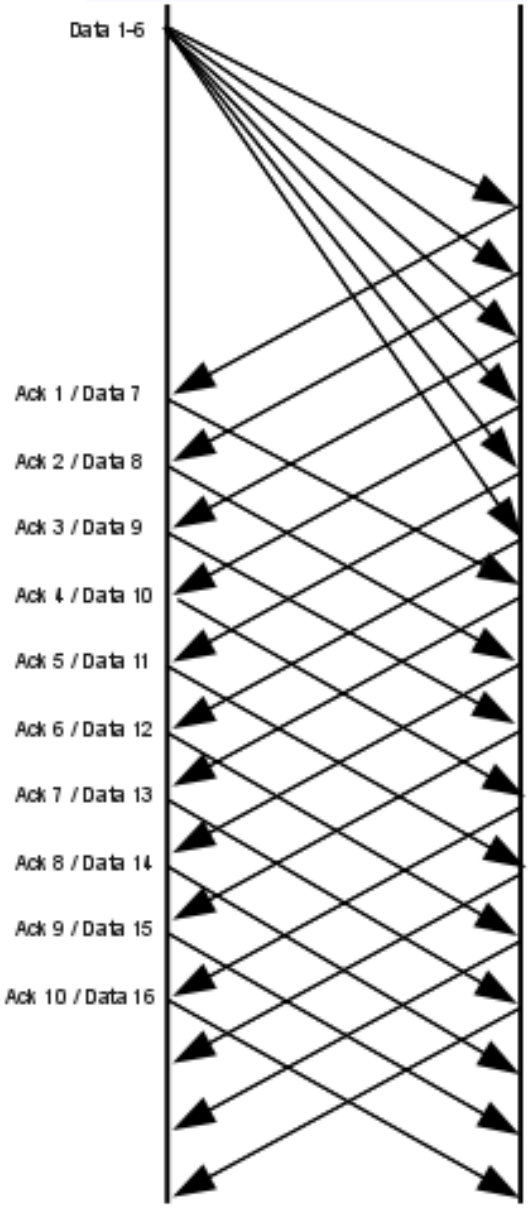
Basic Sliding-Window Concept

- The receiver allocates buffer space for **W frames**.
- Therefore:
  - The sender may transmit up to **W frames without waiting for acknowledgments**.
- **Window size:**

  W = maximum number of outstanding frames



WinSize = 4



WinSize = 6

Sequence Numbers

- Frames are identified using **sequence numbers**.
  - Example:

    0 1 2 3 4 5 6 7

  - After the maximum value, numbering wraps around:

    ... 5 → 6 → 7 → 0 → 1 → 2 ...

- For a **k-bit sequence number**:

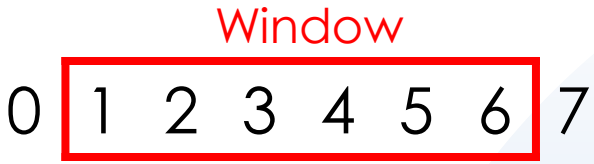
  Sequence *numbers* = 0 *through* $2^k - 1$

Why?

- The sender and receiver need to identify:
  - Which frames have been transmitted
  - Which frames have been acknowledged
  - Which frame is expected next

The Sliding Window

Example with sequence numbers:



- The window represents the frames that the sender is currently allowed to transmit.

As frames are transmitted



- Window moves/shrinks from the trailing edge.

As ACKs arrive

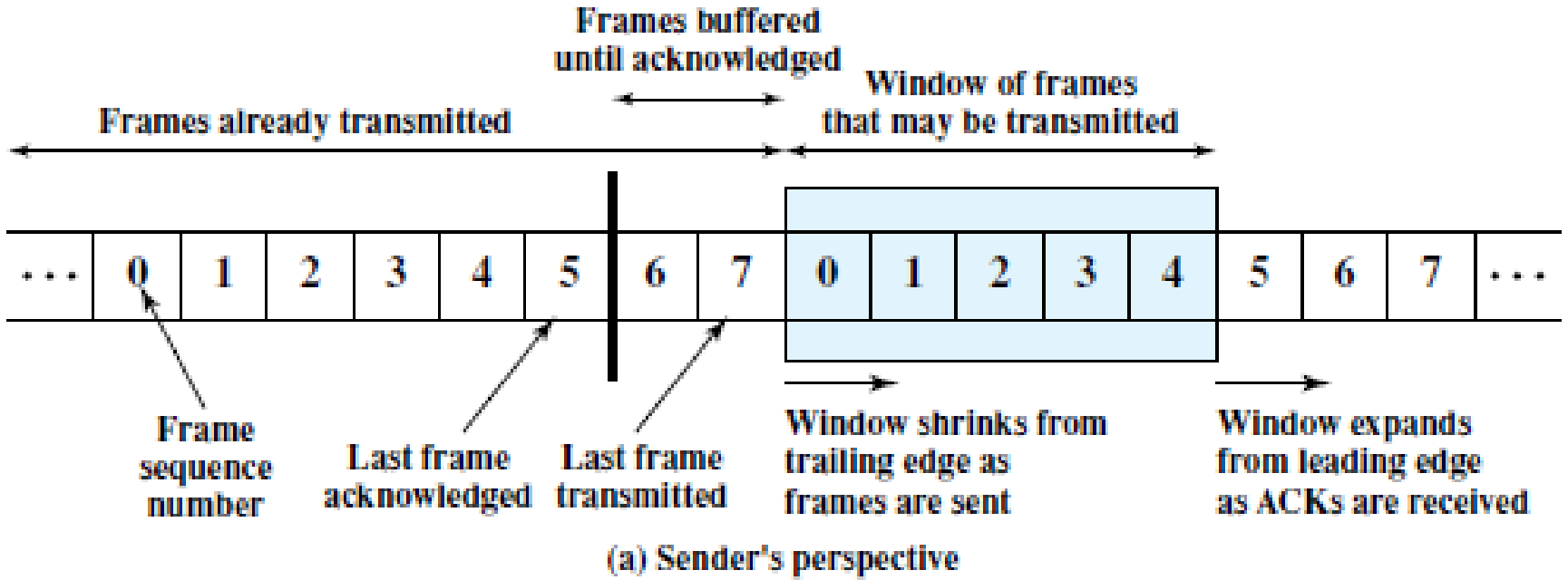


- Window moves/expands from the leading edge.

Sender's Perspective

The sender maintains a window containing frames that:

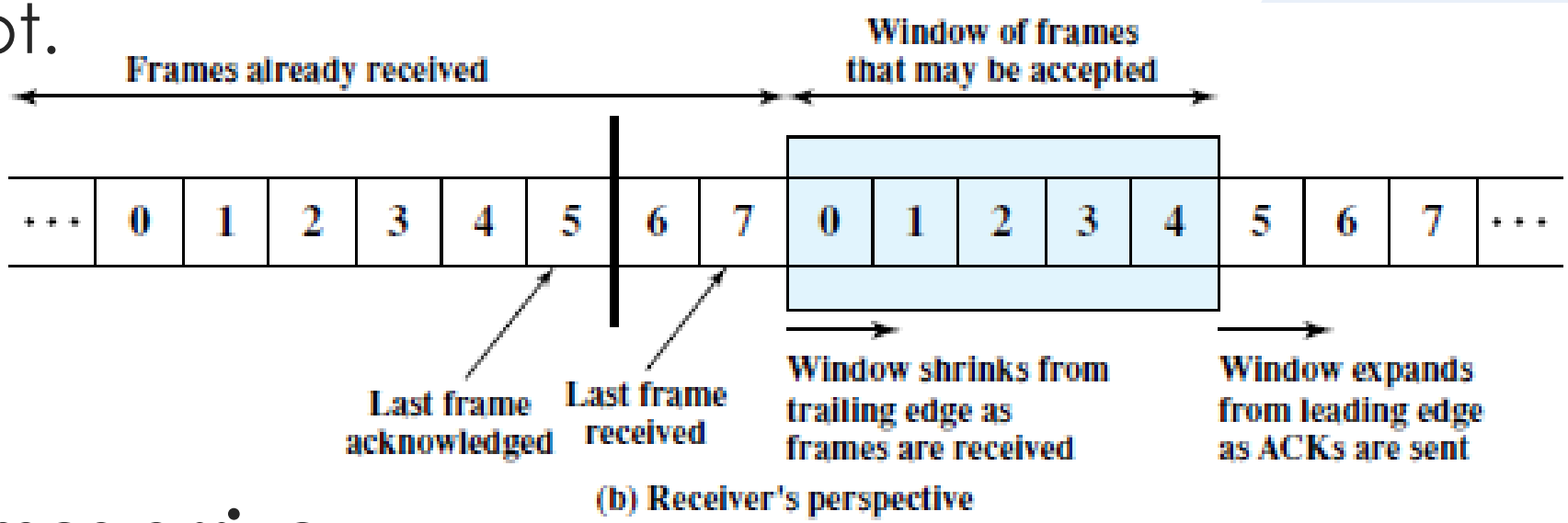
- May be transmitted
- Have not yet been acknowledged
- Must be buffered after transmission



(a) Sender's perspective

Receiver's Perspective

- The receiver also maintains a window.
- It represents the frames that the receiver is prepared to accept.



- **As frames arrive**
  - Receiver window moves forward.
  - Receiver sends acknowledgments.
  - The sender can then advance its window.

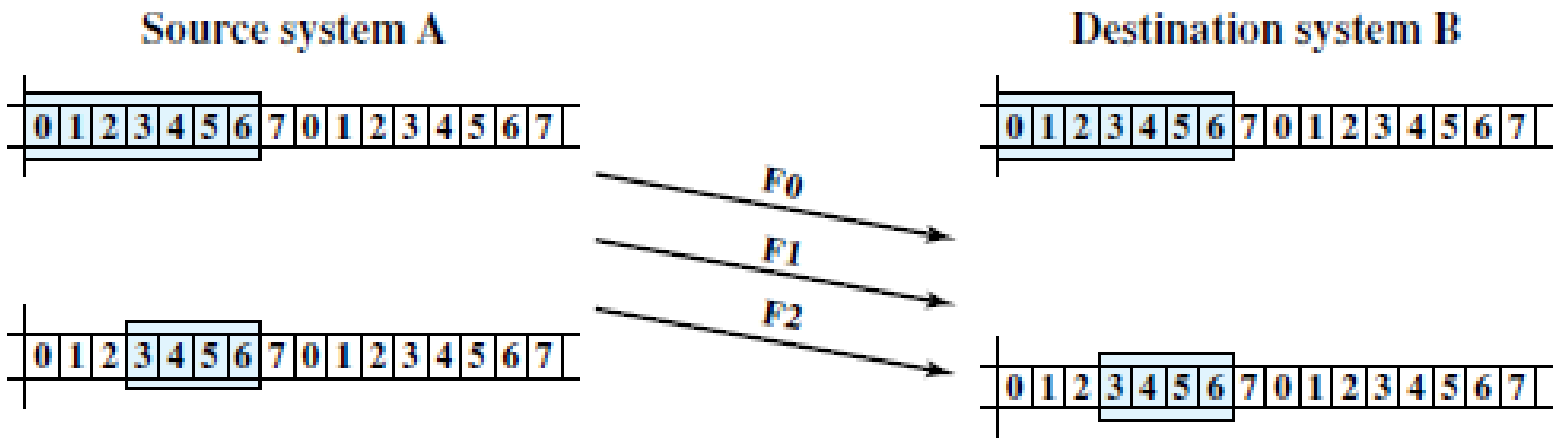
Sliding Window: Step-by-Step Example

Assume:

- 3-bit sequence numbers
- Window size = 7

Initially:

- Window= 0 1 2 3 4 5 6



Sender transmits: *F0 F1 F2*

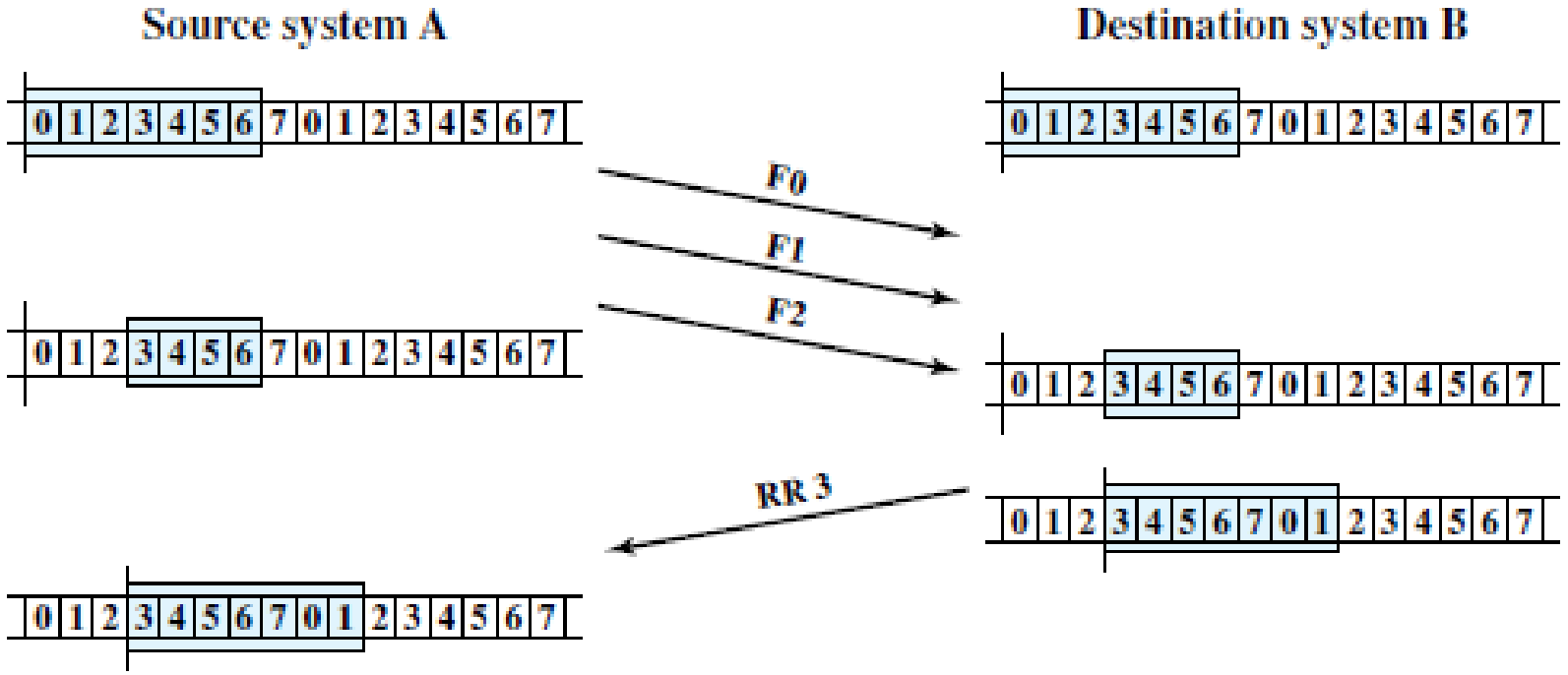
Now:

- Frames sent but not ACKed: *F0 F1 F2*
- Available window moves to: *3 4 5 6*

Sliding Window: Step-by-Step Example

When ACK arrives

- Suppose receiver sends: **RR 3**
- Meaning:
  - Frames through **2** have been received; frame **3** is expected next.
- The sender's window advances.

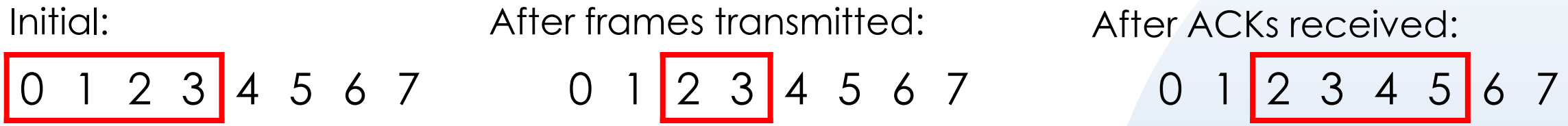


Acknowledgment in Sliding Window

- An acknowledgment can acknowledge **multiple frames**.
  - Example:
    - Receiver receives: *F2 F3 F4*
    - Instead of acknowledging each individually, it can send: *ACK 5*
- Meaning:
  - Frames through **4** have been received; frame **5** is expected next.
- **Benefit**
  - **One acknowledgment can confirm multiple frames.**

Window Movement

The window **slides** as communication progresses.



Two movements

- **Frames transmitted**
  - trailing edge moves forward
- **ACKs received**
  - leading edge moves forward
- **Key idea**
  - The sender can continue transmitting as long as there are positions available inside the window.

10.4
Comparing the Two



Stop-and-Wait vs Sliding Window

Feature	Stop-and-Wait	Sliding Window
Frames in transit	One	Multiple
ACK requirement	Wait before next frame	ACKs can arrive while sending
Receiver buffering	Limited requirement	Buffer space for window
Link utilization	Lower	Higher
Implementation	Simple	More complex
Long-delay links	Inefficient	More efficient
Sequence numbers	Not needed for basic flow control	Used to track frames

Why Sliding Window Is More Efficient

Stop-and-Wait

FRAME 1 ➡️ WAIT ➡️ ACK ➡️ FRAME 2

- Large portions of time may be idle.

Sliding Window

F1 | F2 | F3 | F4 ➡️ ACK ➡️ F5

- Transmission can continue while earlier frames are travelling through the link.

Key idea:

- **Multiple outstanding frames keep the transmission pipeline busy.**

Window Size and Performance

The window size w affects how effectively the link is used.

If: *W is too small*

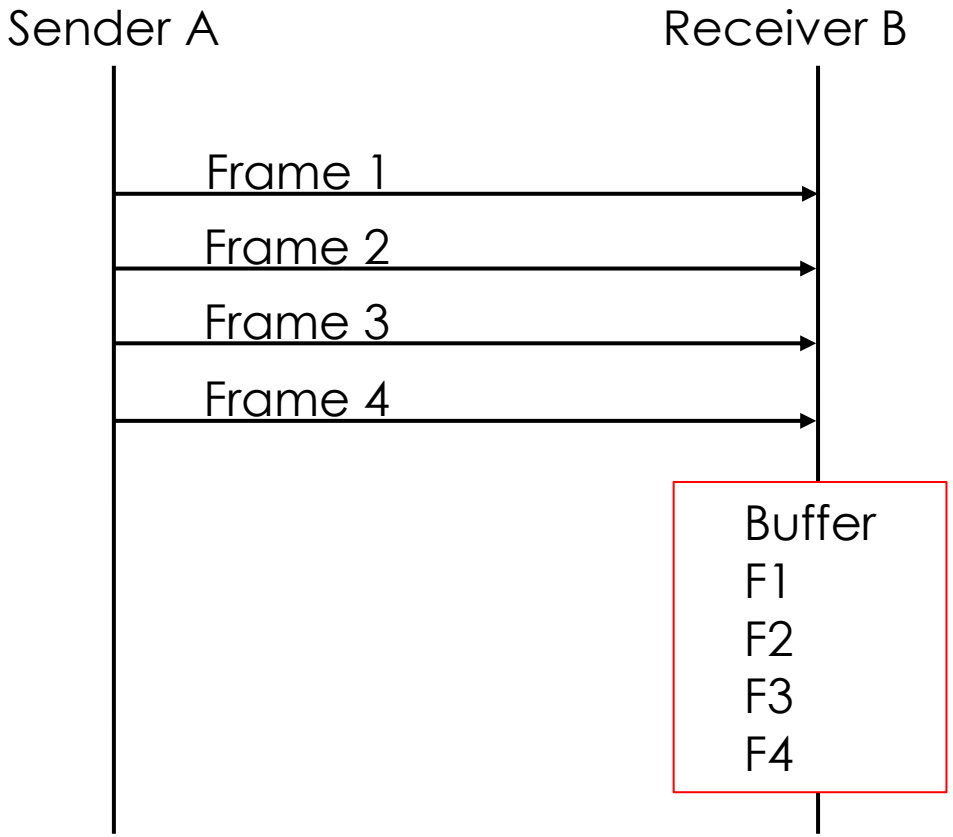
- Sender may have to stop and wait.
- Concept: F1 ➡️ WAIT ➡️ ACK ➡️ F2

If: *W is sufficiently large*

- Sender can keep transmitting while waiting for ACKs.
- Concept: F1 | F2 | F3 | F4 ➡️ ACKs return

Sliding Window and Receiver Buffer

- The receiver must be able to accommodate the frames permitted by the window.



Therefore

- **Window size is related to receiver buffering capacity.**

10.5
An Additional Concept

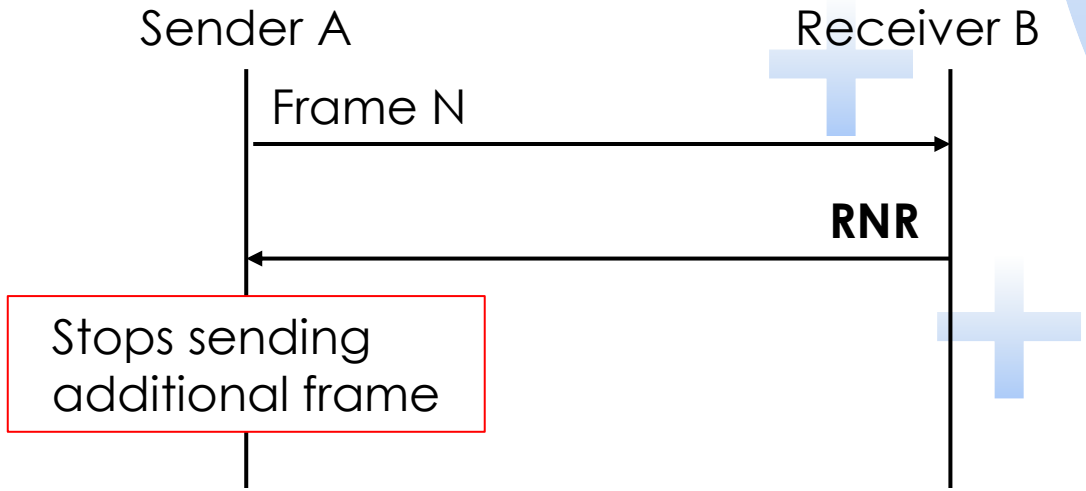


Receive Not Ready (RNR)

- In data-link protocols using sliding-window flow control, the receiver may need to temporarily stop incoming frames.

RNR — Receive Not Ready

- Acknowledges previously received frames.
- Tells the sender **not to send additional frames**.
- Used when the receiver cannot currently accept more frames.



10.6
Flow Control
vs
Error Control



Do Not Confuse Flow Control with Error Control

Flow Control

- **Question:** How fast should the sender transmit?
- Purpose:
  - Prevent receiver overload
  - Manage receiver buffering

Error Control

- **Question:** What should happen when a frame is lost or damaged?
- Purpose:
  - Detect/recover from transmission errors
  - Retransmit when necessary
- **Important**
  - **Stop-and-Wait** and **Sliding Window** are primarily **flow-control techniques**.
  - They can also be used as the basis for error-control mechanisms such as ARQ.

Key takeaways

Stop-and-Wait

- Send one frame.
- Wait for acknowledgment.
- Then send the next frame.
- Simple but can leave the link idle.
- Particularly inefficient when propagation delay is large.

Sliding Window

- Multiple frames can be in transit.
- Frames use sequence numbers.
- Sender maintains a transmission window.
- Receiver maintains a receiving window.
- ACKs cause the window to advance.
- Better line utilization.

Review Questions

1. What is flow control?
2. Why is flow control required between a sender and receiver?
3. What problem can occur if the sender transmits faster than the receiver can process data?
4. What is the purpose of the receiver's buffer?
5. What are the two flow-control techniques discussed in this lecture?

Stop-and-Wait

6. Explain the operation of stop-and-wait flow control.
7. Why can only one frame be in transit in stop-and-wait?
8. What does the sender do after transmitting a frame?
9. Why does stop-and-wait become inefficient on long-distance links?
10. What happens to link utilization when propagation delay increases?

Review Questions

11. What does the following parameter represent?

$$a = \frac{t_{prop}}{t_{frame}}$$

12. Using the following equation calculate the utilization when $a = 1$.

$$U = \frac{1}{1 + 2a}$$

13. What happens to utilization when a becomes very large?

Sliding Window

14. What is sliding-window flow control?
15. Why does sliding-window flow control improve link utilization?
16. What is meant by the **window size** W?
17. Why does the sender need sequence numbers?
18. What happens to the sender's window when a frame is transmitted?

Review Questions

19. What happens to the sender's window when an acknowledgment is received?
20. Why must the sender buffer frames that have been transmitted but not yet acknowledged?
21. What does an acknowledgment with sequence number 5 indicate in the example where frames 2, 3 and 4 have been received?
22. If a 3-bit sequence number is used, what sequence numbers are possible?
23. What happens after sequence number 7?What is the purpose of the receiver's window?
24. What is an RNR message?

Review Questions

Comparison / Application

25. Compare Stop-and-Wait and Sliding-Window flow control.
26. Which technique provides better utilization on a high-delay link? Why?
27. A sender is connected to a receiver through a satellite link. Which flow-control technique would be more efficient? Explain.
28. Why does increasing the window size generally allow better utilization?
29. What happens if the sender is allowed to transmit more frames than the receiver can buffer?
30. Draw a timing diagram showing Stop-and-Wait flow control.
31. Draw a timing diagram showing Sliding-Window flow control with a window size of 4.
32. Explain why sliding-window flow control is called a **"sliding window."**

Review Questions

33. **Scenario:**

A link has a frame transmission time of **10 ms** and a propagation time of **30 ms**. Compare the behavior of Stop-and-Wait with a sliding window that allows multiple frames to be transmitted.

Find:

i. What is a?

ii. What is the approximate Stop-and-Wait utilization?

iii. Why can a larger sliding window improve utilization?

iv. Draw both timing diagrams.

Thank you

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